

基于混合检索器的多模态生成模型研究与实现

中期答辩

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研究目标

- ▶ 面向多模态检索增强生成 (mRAG) 场景，提升检索器返回有效视觉证据的能力。
- ▶ 基于 MRAG-Bench 构建统一评测流程，回答“更懂指令的检索器是否真的能提升下游问答效果”。
- ▶ 以 MagicLens 为核心检索器，对比 GT Oracle、Random、Text-only 与 corpus 检索等策略。

挑战

- ▶ 多模态问答中，性能瓶颈究竟在“候选从哪里来”，还是“候选如何排序”。
- ▶ 当问题真正依赖视觉证据时，现有检索器是否能稳定召回并利用正确图像。

研究动机

- ▶ LLM/MLLM 仍存在知识冻结与幻觉问题，RAG 通过外部知识缓解真实性问题。
- ▶ 在多模态场景下，错误不仅是“文本事实编造”，还包括“视觉事实编造”。
- ▶ 因此需要从文本 RAG 走向 mRAG，并重点提升视觉证据检索质量。

研究意义

- ▶ 为具身智能提供可检索的长期记忆与证据链。
- ▶ 为脑机接口与高可靠辅助沟通提供更低幻觉的检索基础。
- ▶ 为后续检索器训练与扩散模型驱动增强提供实验基础。

mRAG 研究主线

- 框架层：mRAG 从静态单轮走向动态、按需、多轮检索，核心关注幻觉抑制与时效性提升。
- 表示层：CLIP、ALIGN 等统一语义空间方法推动跨模态检索成为可能。
- 检索层：FG-CLIP、Long-CLIP、MagicLens 等工作分别从细粒度理解、长文本建模、开放式指令检索改进效果。

目前仍存在的空缺

- 现有研究更关注通用框架综述，缺少面向“视觉证据是否真正被找对”的系统评测。
- 多数实验没有拆开粗召回与重排两类能力，难以定位 mRAG 的真实瓶颈。



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MRAG-BENCH: VISION-CENTRIC EVALUATION FOR RETRIEVAL-AUGMENTED MULTIMODAL MODELS

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<https://mragebench.github.io>

ABSTRACT

Existing multimodal retrieval benchmarks primarily focus on evaluating whether models can retrieve and utilize external *textual knowledge* for question answering. However, there are scenarios where retrieving visual information is either more beneficial or easier to access than textual data. In this paper, we introduce a multimodal retrieval-augmented generation benchmark, MRAG-BENCH, in which we systematically identify and categorize scenarios where visually augmented knowledge is better than textual knowledge, for instance, more images from varying viewpoints. MRAG-BENCH consists of 16,130 images and 1,353 human-annotated multiple-choice questions across 9 distinct scenarios. With MRAG-BENCH, we conduct an evaluation of 10 open-source and 4 proprietary large vision-language models (LVLMs). Our results show that all LVLMs exhibit greater improvements when augmented with images compared to textual knowledge, confirming that MRAG-BENCH is vision-centric. Additionally, we conduct extensive analysis with MRAG-BENCH, which offers valuable insights into retrieval-augmented LVLMs. Notably, the top-performing model, GPT-4o, faces challenges in effectively leveraging retrieved knowledge, achieving only a 5.82% improvement with ground-truth information, in contrast to a 33.16% improvement observed in human participants. These findings highlight the importance of MRAG-BENCH in encouraging the community to enhance LVLMs' ability to utilize retrieved visual knowledge more effectively.

出发点

- ▶ MRAG-Bench 是 vision-centric benchmark。
- ▶ 核心问题不是“能否检索文本知识”，而是“视觉证据是否比文本更关键”。
- ▶ 这与本文从图像检索器切入 mRAG 的研究目标直接一致。

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Statistic	Number
Total questions	1,353
- Multiple-choice questions	1,353 (100%)
- Questions newly annotated	1,353 (100%)
Total Scenarios	9
Unique number of questions	375
Unique number of answers	463
Total number of images	16,130
Unique number of images	16,130
Human selected images	9,673
Average image size (px)	1076 x 851
Maximum question length	20
Maximum answer length	9
Average question length	8.03
Average answer length	2.16
Average choice number	4

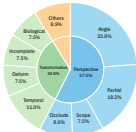


Table 2: Key statistics of MRAG-BENCH. Figure 2: Scenarios distribution of MRAG-BENCH.

Notably, we observe while all models improve with GT knowledge, only proprietary models are able to effectively utilize noisy retrieved multimodal knowledge. This indicates the gap between open-source and close-source models still exists. Open-source models are falling short on their parametric knowledge and the ability to distinguish between high-quality and poor-quality retrieved visually augmented examples. In comparison to humans, GPT-4o achieves only a 5.82% improvement when augmented with GT knowledge and 0.28% with retrieved knowledge, whereas humans demonstrate a 33.16% and 22.91% improvement, respectively. These results highlight the importance of MRAG-BENCH in encouraging the community to develop LVLMs better utilizing of visually augmented knowledge.

2 MRAG-BENCH

2.1 BENCHMARK OVERVIEW

Our benchmark is designed for systematic evaluation of LVLm’s vision-centric multimodal RAG abilities. To achieve this, we focus on evaluating the model’s understanding of image objects that are not commonly associated with its knowledge base, while the collected ground-truth images can help incentivize specific visual concepts within LVLm’s memory. Therefore, we divide our benchmark into two main aspects, as illustrated in the examples in Figure 1:

- *perspective*, refers to the challenges in visual recognition and reasoning that arise when a visual entity is presented from varying viewpoints, scopes, or levels of visibility.
- *transformative*, refers to the challenges that arise when a visual entity undergoes fine-grained physical transformations, making it unfamiliar or not easily associated with the model’s prior knowledge.

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Figure 3: Qualitative examples on MRAG-BENCH. For each scenario, we show the result of GPT-4o (OpenAI, 2023), Gemini Pro (Team et al., 2023), LLaVA-Next-Interleave (Li et al., 2024b) and Mantis-8B-Siglip (Jiang et al., 2024a). The ground-truth answer is in blue.

2.2 BENCHMARK COMPOSITION

MRAG-BENCH provides a systematic evaluation across 9 distinctive multimodal RAG scenarios, with four scenarios focused on the perspective understanding of visual entities, four on transformative understanding, and one categorized as “others”. As illustrated in Figure 2, each scenario comprises 7.5% to 23.8% of the whole benchmark. The selected examples of each scenario is shown in Figure 3. The details of each scenario are introduced as follows.

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研究内容

1. 基于 MRAG-Bench 重构视觉证据导向的 mRAG 评测任务，将检索阶段作为核心变量。
2. 引入 MagicLens 作为多模态检索器，研究其在“关系驱动证据检索”中的实际价值。
3. 设计 GT Oracle、Random、Retrieved-Rerank、Corpus-CLIP 等对照实验，量化不同检索策略的收益与上限。

组织思路

- ▶ 从整体准确率、分场景准确率、GT 覆盖率与失败模式四个层面分析检索器表现。
- ▶ 将实验明确拆为 **Rerank line** 和 **Corpus line**。
- ▶ 分别回答排序收益与粗召回收益。



为什么引入 MagicLens

- ▶ 传统图像检索更偏向视觉相似性，难表达“内部视角”“相关物品”“不同时刻”等开放关系。
- ▶ MagicLens 将检索定义为 $(image_q, instruction) \rightarrow image_t$ ，强调关系驱动的目标搜索。
- ▶ 这与 MRAG-Bench 中大量依赖视觉证据与跨图关系的题型高度契合。

论文要点

- ▶ 开放式 instruction 让检索从“像不像”转向“关系是否匹配”。
- ▶ MagicLens 在 多类检索 benchmark 上表现稳定，同时保持较好的参数效率。

arXiv:2403.19651v2 [cs.CV] 24 Jun 2024

MagicLens: Self-Supervised Image Retrieval with Open-Ended Instructions

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Abstract

Image retrieval, i.e., finding desired images given a reference image, inherently encompasses rich, multi-faceted search intents that are difficult to capture solely using image-based measures. Recent works leverage text instructions to allow users to more freely express their search intents. However, they primarily focus on image pairs that are visually similar and/or can be characterized by a small set of pre-defined relations. The core thesis of this paper is that text instructions can enable retrieving images with richer relations beyond visual similarity. To show this, we introduce MagicLens, a series of self-supervised image retrieval models that support open-ended instructions. MagicLens is built on a key novel insight: image pairs that naturally occur on the same web page contain a wide range of implicit relations (e.g., *location view of*), and we can bring these implicit relations explicit by synthesizing instructions via foundation models. Trained on 36.7M (query image, instruction, target image) triples with rich semantic relations mined from the web, MagicLens achieves results comparable with or better than prior best on eight benchmarks of various image retrieval tasks, while maintaining high parameter efficiency with a significantly smaller model size. Additional human analyses on a 1-OM image dataset corpus further demonstrate the diversity of search intents supported by MagicLens. Code and models are publicly available at the Project Website.

1. Introduction

Image retrieval is a long-established problem in computer vision (Efthys et al., 2009; Gordo et al., 2016) with a wide

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Figure 1. Top-1 retrieved images using MagicLens and the prior state-of-the-art (SOTA) method (Gu et al., 2023) from a retrieval pool with 1-OM images. The prior SOTA method, while accepting text instructions, primarily retrieves images based on visual similarity to the query image, ignoring the nuances of the text instructions. In contrast, MagicLens excels at retrieving both visually similar images and those that align with the deeper meaning and context of the text instructions — even when the images do not resemble the query. For example, if given a query image of the Burj Al Arab and the instruction “Find other attractions in this country”, it can successfully locate images of the Palm Jumeirah in Dubai.

range of real-world applications, such as visual search, object localization, and re-identification. However, since its inception, this task has suffered from ambiguous definitions due to the complex and rich content encapsulated in images. Similar images may differ in key aspects, and different images can share commonalities. In image search scenarios, users frequently present multiple search intents for a single query image, indicating that more image relevance is insufficient for precise search results. For instance, when searching with an image of the Burj Al Arab in Dubai, i.e., Dubai 1 (see Figure 1), a user might seek other attractions in Dubai or an interior view, each relating differently to the query image. Therefore, incorporating text instructions that articulate search intents is essential and indispensable for enhancing retrieval accuracy. Ideally, models should accurately capture and interpret diverse real-world search intents as conveyed by open-ended text instructions.

These open-ended search instructions, upon a wide range of topics and concepts, reflect the diverse ways users



自监督数据构造

- 从同一网页中挖掘自然共现图像对，而不是只依赖人工定义关系。
- 用 LMM/LLM 为图像对生成开放式 instruction，形成 (query image, instruction, target image) 三元组。
- 最终构建约 3670 万条训练样本，覆盖大量隐式语义关系。

模型结构

- 共享参数的 dual-encoder，以 CLIP/CoCa 为初始化。
- 通过 self-attention 融合图像与指令，再用 attention pooling 压缩成单向量表示。
- target 端使用 ($image_t$, "") 编码，并采用 contrastive loss；同时把 query image 本身作为 hard negative。

MagicLens: Self-Supervised Image Retrieval with Open-Ended Instructions

PaLI (Chen et al., 2023a) to generate captions. Each type of metadata provides textual information about images from different perspectives. See Appendix A for more details.

(f) *Scoring & Filtering.* After obtaining groups of images along with their extensive metadata, we pair up images within the same group and eliminate unsuitable pairs using a combination of relevance measures. We use the CLIP image-to-image score to assess visual relevance and the text-to-text score for non-visual relevance. Image pairs that don't meet the criteria, such as those with low scores in both aspects, are excluded from consideration. To avoid the over-sampling of redundant images and duplicated relations, we set a maximum of three pairs for each group, thereby ensuring a more uniform distribution of images and relations in our training data (see Figure 5).

Open-Ended Instruction Generation. With the informative metadata of high-quality paired images, LLMs are able to well understand the image content (ICA and caption) and their background information (Alt-text). Using instruction (Chang et al., 2023), few-shot demonstrations (Brown et al., 2020), and chain-of-thought prompting (Wei et al., 2022) techniques, PaLM2 (Anil et al., 2023) generates open-ended instructions that precisely connect the paired images ($image_q, image_t$). Figure 3 illustrates generated instructions and Appendix 10 shows the detailed prompt and demonstrations. Eventually, we obtain 36.7M examples ($image_q, inst, image_t$) for self-supervised image retrieval training.

3.2 MagicLens Model

Model Design. As shown in Figure 4, we adopt a simple dual-encoder architecture with shared parameters and initialize the backbone vision and language encoders with CoCa (Yu et al., 2022) or CLIP (Radford et al., 2021). To enable deep modality integration, we introduce multiple self-attention layers and design a single multi-head attention pooler, compressing the multimodal inputs into a single embedding e for later matching. Additionally, since the retrieval target comprises only an image without accompanying text, we employ an empty text entry "" to transform the target into a multimodal input. We denote e_q as the embedding for the multimodal query ($image_q, inst$) and e_t as the embedding for the target ($image_t, ""$). Considering the efficiency, we propose MagicLens-R and MagicLens-L, initialized with the base and large checkpoints, respectively.

Model Training. We use a simple contrastive loss to train MagicLens. Our model is updated by contrasting the paired query-target against other targets in one training batch. In particular, as the query image itself ($image_q$) can be a challenging hard negative for the multimodal query ($image_q, inst$), we combine the query image itself and an empty text to encode ($image_q, ""$) to get e_q' as an additional query neg-

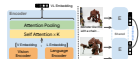


Figure 4. Model architecture and training of MagicLens Encoder (E), which takes the vision and language embeddings and fuses them as a sequence to self-attention layers for modality integration.

ative example. To scale up the number of negative examples, for each query images, we use all query negatives and other target negatives in the same batch. Formally, for the i -th training example, the loss function $\mathcal{L}_i$ is defined as,

$$-\log \frac{\exp(\text{sim}(e_q, e_t))}{\exp(\text{sim}(e_q, e_t)) + \sum_{j \neq t} \exp(\text{sim}(e_q, e_j)) + \sum_{j \neq q} \exp(\text{sim}(e_q', e_j))}$$

where $\text{sim}(\cdot)$ indicates a cosine similarity function $\frac{e_i \cdot e_j}{\|e_i\| \|e_j\|}$. N refers to the sampled batch size, and τ is a temperature hyperparameter for logit scaling. Please refer to Appendix A for more implementation details.

4. Experiments

4.1. Experiment Setup

Benchmarks and Metrics. To comprehensively evaluate MagicLens' multimodality-to-image retrieval ability, we consider three related tasks in a zero-shot, one-checkpoint setting: (1) composed image retrieval (CIR), (2) domain transfer retrieval, and (3) conditional image similarity. Each task has different yet limited sets of image relations. Table 2 shows the detailed statistics of five benchmarks.

Composed Image Retrieval. We consider one domain-specific and two open-domain benchmarks to evaluate the model's domain adaptability and its capability on real-world natural images, respectively. FTQ (Wu et al., 2023) is a fashion-domain benchmark with three disjoint retrieval sub-tasks: dress, shirt, and topset. Following previous work (Saito et al., 2023; Baldrati et al., 2023; Gu et al., 2024), we evaluate on its validation set and report recall averaged over sub-tasks. $CLIP$ (Gua et al., 2023) is the first dataset constructed on natural images (Suh et al., 2019) with nine pre-defined relations between the query and the target images. It also includes a subset retrieval setting where models retrieve a target image from a dedicated small subset for each query. However, in addition to the limited size of the retrieval pool, it also suffers from false negative issues, as pointed out by Baldrati et al. (2023). We utilize recall (R and R_{sub}) to evaluate standard retrieval and subset retrieval. In contrast, to better align with the real-world large-scale

统一评测流程

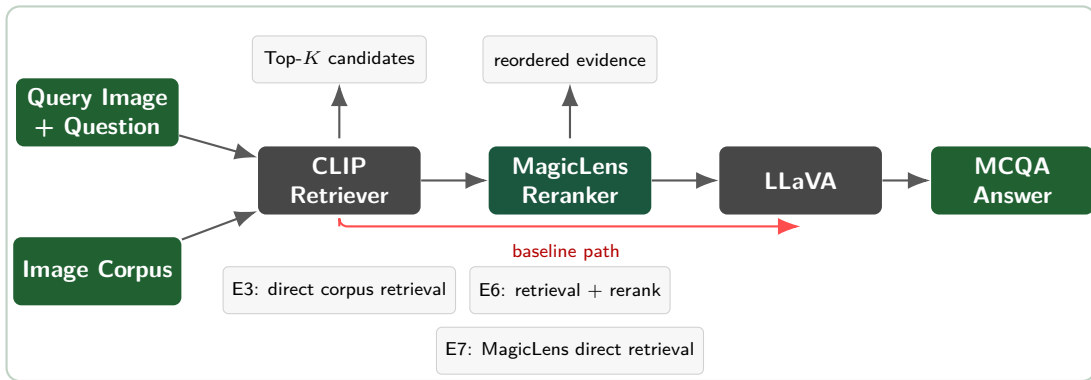
1. 输入 query image + question
2. 检索 Top- k 候选图像
3. 将问题与候选图送入 LLaVA
4. 输出多项选择答案并统计指标

对照设置

- ▶ E0: LLaVA + GT-RAG baseline
- ▶ E1/E2: 固定候选集上的 MagicLens rerank
- ▶ E3: CLIP 从完整 corpus 召回 Top-5
- ▶ E6: CLIP 粗召回 + MagicLens 重排
- ▶ E7: MagicLens 直接从完整 corpus 召回

评价指标

整体 Accuracy、九类场景分布、GT 覆盖率、接近 Oracle 的程度，以及“检索对但生成错 / 检索错但生成对”的失败模式。



E3

CLIP 直接从完整 corpus 召回 Top- K 候选。

E6

CLIP 粗召回后，再用 MagicLens 重排。

E7

MagicLens 直接作为全库 retriever。



任务定义

$$\begin{aligned} Q &= (x_q, q) \\ I &= R(Q) = \{i_1, \dots, i_K\} \\ \hat{y} &= M(Q, I) \end{aligned}$$

排序分数

$$s_i = \langle z_q, z_i \rangle$$

排序损失

$$\mathcal{L}_{\text{rank}} = -\log \frac{\exp(s(q, i^+))}{\sum_j \exp(s(q, i_j))}$$

联合优化目标

$$\begin{aligned} \mathcal{L}_{\text{ans}} &= -\log p(\hat{y} = y^* \mid Q, I) \\ \mathcal{L} &= \mathcal{L}_{\text{rank}} + \lambda \mathcal{L}_{\text{ans}} \end{aligned}$$

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已完成

- ▶ 搭建 MRAG-Bench 数据与评测管线，明确图像候选进入模型的真实规则。
- ▶ 跑通 GT-RAG、Retrieved-Rerank、CLIP corpus Top-5 等基线实验。
- ▶ 完成候选重叠统计与问题拆分，明确区分 rerank 与 corpus 两条实验线。

正在进行

- ▶ 集成 MagicLens 到完整 corpus 检索流程，形成 E6 主实验。
- ▶ 新增 E7 直检索实验脚本，用于评估 MagicLens 作为 direct retriever 的效果。
- ▶ 分场景分析检索收益，定位 Temporal、Scope、Incomplete 等难点场景。



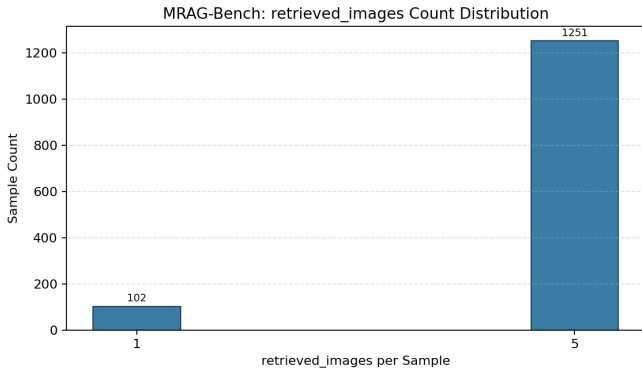
MRAG-Bench 数据特征

- ▶ 共 1353 个测试样本，覆盖 9 类视觉证据依赖场景。
- ▶ GT 候选平均 4.70 张，retrieved_images 原始字段固定为 5 张。
- ▶ Incomplete 场景在 dataloader 中会被截断为 1 张，需单独分析。

重叠统计结论

- ▶ 44.12% 样本中，retrieved 与 GT 完全无重叠。
- ▶ 平均 GT 覆盖率仅 22.78%。
- ▶ 说明数据集自带候选与 oracle 候选存在显著偏差。

Source	1 张	5 张	Avg.
GT raw	102	1251	4.70
Retrieved raw	0	1353	5.00
Retrieved effective	102	1251	4.70



观察

- ▶ 数据集给定候选与 GT 候选分布并不一致。
- ▶ 一部分样本中，retrieved 候选几乎不覆盖真正有效证据。
- ▶ 这也是本文把局部 rerank 与全库检索分开讨论的直接原因。



实验线含义

- ▶ E0: GT-RAG, 上界参考。
- ▶ E1: GT 候选 + MagicLens rerank, 测排序收益。
- ▶ E2: 数据集候选 + MagicLens rerank, 测局部重排能力。
- ▶ E3: CLIP 全库直检索, 测粗召回 baseline。
- ▶ E4: No-RAG, 测无外部证据基线。
- ▶ E5: GT 候选但关闭 rerank, 用于隔离 E1 的排序收益。
- ▶ E6: CLIP 粗召回 + MagicLens rerank, 测混合检索。
- ▶ E7: MagicLens 全库直检索, 测 direct

指标在回答什么

- ▶ Overall Accuracy: 最终问答效果。
- ▶ 分场景 Accuracy: 不同视觉证据类型下的稳定性。
- ▶ E1 vs E5: 判断 rerank 是否真正带来收益。
- ▶ E3 vs E6: 判断重排能否改善 CLIP 粗召回。
- ▶ E3 vs E7: 判断 MagicLens 是否适合直接做全库召回。
- ▶ E4 vs 其余实验: 判断“给证据”是否真的优于“不给证据”。



Track	Method	Overall (%)
Control	E0: GT-RAG	59.05
Rerank	E1: GT+ML rerank	60.16
Rerank	E2: Ret+ML rerank	51.74
Corpus	E3: CLIP corpus	50.41
Control	E4: No-RAG	53.14
Control	E5: GT no rerank	60.16
Corpus	E6: CLIP+ML rerank	50.33
Corpus	E7: ML corpus	47.97

初步观察

- ▶ E1>E5 不明显：GT 条件下 rerank 收益有限。
- ▶ E4>E3/E6/E7：噪声证据可能伤害回答。
- ▶ E6≈E3：当前重排尚未稳定改善粗召回。
- ▶ E7<E3：MagicLens 直检索仍弱于 CLIP baseline。

分场景现象

E4 在 Incomplete 上达到 33.33%，高于 E3 的 22.55%；E6 在 Scope (55.88%) 和 Incomplete (27.45%) 上高于 E3，但在 Partial、Occlusion 与 Biological 上下降，说明当前提升并不稳定且场景依赖明显。



E7 核心结果

- ▶ 任务：MagicLens 直接从完整 corpus 召回 Top-5。
- ▶ Overall Accuracy: 47.97% (649/1353)。
- ▶ 对比 E3: 低于 CLIP corpus baseline 的 50.41%。

Scenario	E3	E7
Angle	52.48	51.55
Partial	54.07	47.56
Scope	48.04	45.10
Occlusion	54.63	48.15
Temporal	46.31	46.98
Deformation	58.82	50.00
Incomplete	22.55	32.35
Biological	50.00	50.98
Others	57.50	51.67

初步判断

E7 说明 MagicLens 可以直接承担全库召回，但当前整体稳定性仍不如 CLIP；不过它在 Incomplete 场景上有明显优势，说明其在结构缺失类问题上存在潜力。



Control / Rerank

- ▶ E0: LLaVA + GT-RAG baseline, 作为较高上界。
- ▶ E1: 在 GT 候选集上做 MagicLens rerank。
- ▶ E2: 在数据集给定候选上做 MagicLens rerank。
- ▶ E4: LLaVA + No-RAG, 对应无外部知识基线。
- ▶ E5: GT 候选输入, 但关闭 MagicLens 重排, 用于隔离 rerank 收益。

Corpus 线

- ▶ E3: CLIP 从完整 corpus 直接召回 Top-5。
- ▶ E6: CLIP 粗召回 Top-5, 再由 MagicLens 执行 rerank。
- ▶ E7: MagicLens 直接从完整 corpus 中召回 Top-5, 不经过 CLIP 粗召回。

设计目的

通过 E3、E6、E7 的对照, 区分“CLIP 粗召回”“CLIP+MagicLens 混合检索”与“MagicLens 直接召回”三类能力。



Track	Method	Overall	Perspective				Transformative				Others
			Angle	Partial	Scope	Occlusion	Temporal	Deformation	Incomplete	Biological	
Control	E0: GT-RAG	59.05	59.63	64.23	66.67	67.59	57.72	56.86	29.41	55.88	64.17
Rerank	E1: GT+ML rerank	60.16	61.49	65.45	66.67	69.44	58.39	57.84	30.39	55.88	65.00
Rerank	E2: Ret+ML rerank	51.74	51.24	50.41	59.80	59.26	48.99	55.88	33.33	50.00	59.17
Corpus	E3: CLIP corpus	50.41	52.48	54.07	48.04	54.63	46.31	58.82	22.55	50.00	57.50
Control	E4: No-RAG	53.14	56.21	56.10	55.88	57.41	47.65	55.88	33.33	50.98	55.83
Control	E5: GT no rerank	60.16	59.94	66.67	63.73	66.67	61.74	56.86	29.41	57.84	67.50
Corpus	E6: CLIP+ML rerank	50.33	54.04	49.19	55.88	51.85	46.31	56.86	27.45	49.02	56.67
Corpus	E7: ML corpus	47.97	51.55	47.56	45.10	48.15	46.98	50.00	32.35	50.98	51.67

阅读提示

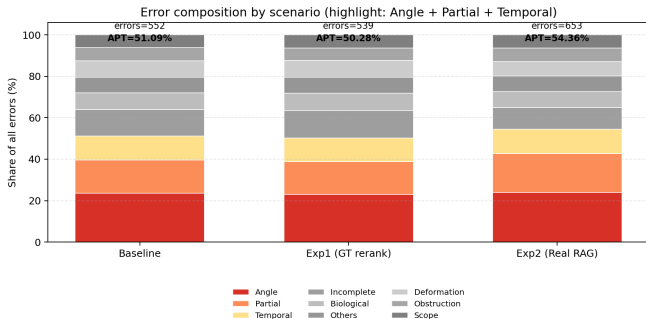
当前 E0-E7 已形成完整对照链。阅读时可优先看三组比较：E1 vs E5（排序收益）、E3 vs E6（混合检索收益）、E3 vs E7（direct retrieval 能力）。

研究进度与结果

研究结果四：错误分布对比



南京师范大学



当前启发

- ▶ 错误并非均匀分布，而是集中出现在若干困难场景。
- ▶ 难点主要落在 Temporal、Scope、Incomplete 等需要强视觉证据的场景。
- ▶ 后续应优先围绕这些场景优化 query 设计、召回策略和重排机制。



阶段结论

- ▶ 仅比较 overall accuracy 不足以解释 mRAG 流程，必须把“候选生成”和“候选排序”分开讨论。
- ▶ MagicLens 在固定候选集上的收益说明其具备一定的排序价值，但这还不等于完整检索能力已经解决。

核心判断

- ▶ 数据集给定候选与 GT 的低重叠率表明，真实 mRAG 的关键问题仍是如何从完整图像库中召回“有用证据”。
- ▶ 从 E1/E5、E3/E6、E3/E7 三组对照看，当前主要矛盾不在“是否加 MagicLens”，而在“给 MagicLens 什么候选、什么 instruction、它究竟做 rerank 还是 direct retriever 更合适”。

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实验计划

1. 完成 E6: 在完整 corpus 上采用 CLIP 粗召回 + MagicLens 重排, 验证主方案收益。
2. 训练轻量级 prompt 生成模型, 为 MagicLens 自动生成更适合检索的 query / instruction。
3. 以“提升 GT 命中率、减少噪声候选、最终提升回答准确率”为目标, 系统优化检索提示词设计。
4. 对 Temporal、Scope、Incomplete 等困难场景开展消融分析, 定位失败原因。

进一步工作

- ▶ 补充 No-RAG、w/o rerank 等控制实验, 形成完整的证据链。
- ▶ 比较人工 prompt、规则模板与小模型生成 prompt 的差异, 分析提示词与检索质量之间的关系。
- ▶ 探索扩散模型驱动的数据增强与检索器训练。
- ▶ 对比 E6 与 E7, 判断 MagicLens 更适合作为 reranker 还是 direct retriever。
- ▶ 为后续具身/脑机接口应用奠定基础。



预期研究结论

- ▶ 证明多模态检索器相较于纯文本或弱检索策略具有明确收益。
- ▶ 定位完整 mRAG 中的主要性能瓶颈，判断其更偏向粗召回还是重排。
- ▶ 识别哪些类型的问题真正依赖视觉证据，形成面向后续优化的场景画像。

预期成果形式

- ▶ 一套可复现的 mRAG 检索评测流程与实验协议。
- ▶ MRAG-Bench 上系统性的检索器对比结果与错误分析。
- ▶ 形成一套面向 MagicLens 的提示词优化方法，为提升 GT 覆盖率、抑制噪声检索和改进最终回答效果提供实验基础。

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- 5. 参考文献

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